

# Panagiotis Theodoropoulos

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## Research Interests

Deep generative models, (Entropic) Optimal Transport, Dynamic Transport maps, Dynamical Systems, Differential Equations, Schrödinger Bridge, Stochastic Optimal Control, Robust Optimal Control, Scalable higher-order optimization, Dynamic Programming, Hamilton-Jacobi Bellman principle.

## Education

Georgia Institute of Technology, Atlanta, GA  
*PhD in Machine Learning* 2022–  
Advisor: [Dr. Evangelos Theodorou](#)

University of Patras, Patras, Greece  
*B.S.- M.S. in Mechanical Engineering* (GPA: 9.00/10.00, Ranked 1st in Class of 2021) 2016–2021  
Advisor: [Dr. Spilios Fassois](#)

## Research Experience

Morgan Stanley, New York, NY  
*Machine Learning Research Associate* (Advisor: [Wei Deng](#)) June–August 2026

Prisma Electronics, Athens, Greece  
*Machine Learning Engineer* (Supervisor: [Christos Spandonidis](#)) 10/2020–06/2022

Prisma Electronics, Athens, Greece  
*Machine Learning Engineer Intern* (Supervisor: [Christos Spandonidis](#)) Summer of 2020

Stochastic Mechanical Systems & Automation, Patras, Greece  
*Undergraduate Research Associate* (Supervisor: [Fassois Spilios](#)) 09/2019 - 07/2021

## Publications

(\*Equal Contribution, <sup>†</sup>Equal Advising)

## Conferences

- [C1] **Theodoropoulos, P.**, Nam, J., Theodorou, E. A., & Choi, J.  
Generalized Schrödinger Bridge on Graphs.  
*International Conference on Machine Learning (ICML) 2026.*
- [C2] **Theodoropoulos, P.**, Saravanos, A. D., Theodorou, E. A.<sup>†</sup>, & Liu, G.-H.<sup>†</sup>  
Momentum Multi-Marginal Schrödinger Bridge Matching.  
Neural Information Processing Systems (NeurIPS) 2025
- [C3] **Theodoropoulos, P.**, Komianos, N., Pacelli, V., Liu, G.-H., & Theodorou, E. A.  
Feedback Schrödinger Bridge Matching. [Oral, top 1.0%]  
*International Conference on Learning Representations (ICLR) 2025.*

- [C4] **Theodoropoulos, P.**, Liu, G.-H., Chen, T., Saravanos, A. D., & Theodorou, E. A.  
A Robust Differential Neural ODE Optimizer.  
*International Conference on Learning Representations (ICLR) 2024.*
- [C5] **Theodoropoulos, P.**, Spandonidis, C., Giordamlis, C. & Fassois, S.  
Stability and Safety of Ships and Ocean Vehicles: Enhancing vessel's operational safety using Convolutional Neural Networks and Big Data techniques.  
*International Conference on the Stability and Safety of Ships and Ocean Vehicles (STAB&S) (2021)*
- [C6] **Theodoropoulos, P.**, Spandonidis, C., Giordamlis, C.  
Combined multi-layered big data and responsible AI techniques for enhanced decision support in Shipping.  
*International Conference on Decision Aid Sciences and Application (DASA) (2020)*

## Journals

- [J1] **Theodoropoulos, P.**, Spandonidis, C. C., & Fassois, S.  
Use of Convolutional Neural Networks for vessel performance optimization and safety enhancement. *Ocean Engineering*, (2022).
- [J2] Spandonidis, C. C.<sup>\*</sup>, **Theodoropoulos, P.**<sup>\*</sup>, Giannopoulos, F., Galiatsatos, N. & Petsa, A.  
Evaluation of deep learning approaches for oil & gas pipeline leak detection using wireless sensor networks. *Engineering Applications of Artificial Intelligence*, (2022).
- [J3] **Theodoropoulos, P.**, Spandonidis, C. C., Giannopoulos, F. & Fassois, S.  
A deep learning-based fault detection model for optimization of shipping operations and enhancement of maritime safety. *Sensors* (2022).
- [J4] **Theodoropoulos, P.**, Spandonidis, C. C., Themelis, N., Giordamlis, C. & Fassois, S.  
Evaluation of Different Deep-Learning Models for the Prediction of a Ship's Propulsion Power. *Journal of Marine Science and Engineering*, (2021).
- [J5] Spandonidis, C., Giannopoulos, F., Sedikos, E., Reppas, D. & **Theodoropoulos, P.**  
Development of a MEMS-based IoV system for augmenting road traffic survey. *IEEE Transactions on Instrumentation and Measurement*, (2022).

## Workshops

- [W1] **Theodoropoulos, P.**, Liu, G.-H., Chen, T. & Theodorou, E.  
Game Theoretic Neural ODE Optimizer.  
In ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems (2023).

## Invited Talks

- **Morgan Stanley ML Research Speaker Series** — *Stochastic Optimal Control and Optimal Transport for Generative Modeling.*

## Honors & Awards

- Awarded by 'Techniko Epimelitirio Elladas - TEE' for graduating first in undergraduate studies
- Gerontelis Foundation Doctoral Scholar

- Salutorian, Graduating Class of 2021, Department of Mechanical Engineering, University of Patras
- Eurobank scholar during undergraduate studies
- Eleytherios Venizelos scholar during undergraduate studies

## Peer Review Service

### Conferences:

- International Conference on Machine Learning (ICML), since 2025
- International Conference on Learning Representations (ICLR), since 2025
- AAAI Conference on Artificial Intelligence (AAAI), since 2026

### Journals:

- Transactions on Machine Learning Research (TMLR), since 2026
- Engineering Applications of Artificial Intelligence
- Ocean Engineering
- Sensors
- International Journal of Maritime Engineering

## Teaching Experience

**Head Teaching Assistant**, Georgia Institute of Technology

Spring 2025

*COE 2001: Statics*

- Led weekly recitations for 50+ students
- Assisted in course coordination, homework grading rubric preparation, and grading.

## Skills

Programming: Python, Matlab, C/C++  
 Packages: Pytorch, Tensorflow, CVXPY